

Optimization

2.3 Successive quadratic interpolation (Powell's method)

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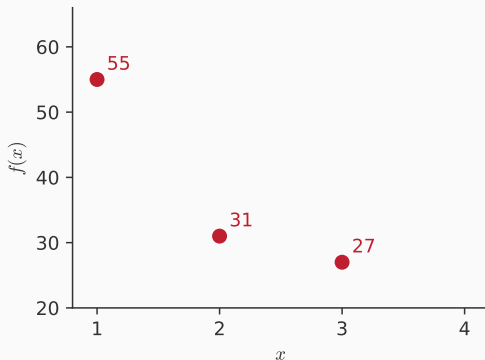
The order or the numbers?

Three evaluations of the guiding example:

$f(1) = 55$, $f(2) = 31$, $f(3) = 27$. Golden section would only use the **order**: $f(1) > f(2) > f(3)$.

But the **numbers** say more: from 1 to 2, f drops by 24; from 2 to 3, by only 4 — f is curving upwards.

Can we use the numbers to guess the shape of f ?



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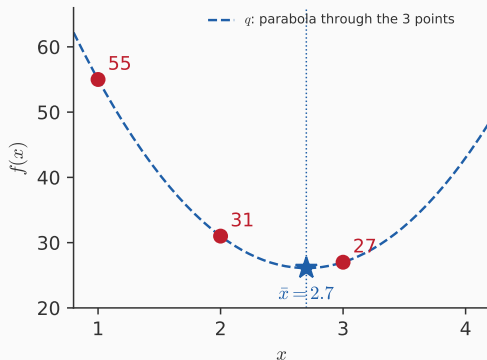
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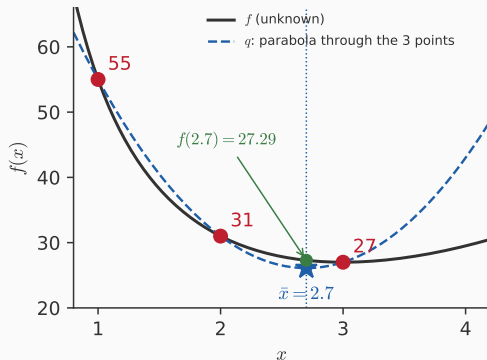
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Can we use the numbers to guess the shape of f ?

Through the three points we pass the simplest model that curves: a **parabola** q .

If q were f , the minimum would be at $\bar{x} = 2.7$.
One new evaluation: $f(2.7) = 27.29$, 0.3 away from the true minimum ($x^* = 3$).



If the parabola were f , where would the minimum be?

Parabola $q(x) = c_0 + c_1x + c_2x^2$ with $q(1) = 55$,

$q(2) = 31$, $q(3) = 27$: $q(x) = 10x^2 - 54x + 99$

(check: $10 - 54 + 99 = 55$; $40 - 108 + 99 = 31$;

$90 - 162 + 99 = 27$).

Minimum of the model: $q'(x) = c_1 + 2c_2x = 0$,

$$\boxed{\bar{x} = -\frac{c_1}{2c_2}} = \frac{54}{20} = 2.7.$$

A minimum because $c_2 = 10 > 0$ (with $c_2 < 0$ it would be a *maximum*).

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$q(2.7) = 26.1$ but $f(2.7) = 27.29$: the model is *not* f , but it points to the right region.

We keep the three best points, 2, 2.7, 3, fit a new parabola, and repeat. If f were a strictly convex quadratic, $q = f$ and $\bar{x} = x^*$ in a single iteration; near the minimum, a smooth function *looks like* a parabola.

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This is the idea of **successive quadratic interpolation** (Powell's method).

Here, "Powell's method" is this *one-dimensional* interpolation; not to be confused with Powell's *multidimensional* derivative-free conjugate-direction method (e.g. `method='Powell'` in SciPy).

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Computing $\bar{x}$ without solving a system

With any three points $x_1 < x_2 < x_3$, q is written in **Newton form** — the coefficients come out one at a time:

$$q(x) = a_0 + a_1(x - x_1) + a_2(x - x_1)(x - x_2),$$
$$a_0 = f_1, \quad a_1 = \frac{f_2 - f_1}{x_2 - x_1}, \quad a_2 = \frac{1}{x_3 - x_2} \left(\frac{f_3 - f_1}{x_3 - x_1} - a_1 \right).$$

- a_1 : slope of the chord between the first two points (first-order divided difference).
- a_2 : second-order divided difference — it measures the **curvature** ($a_2 = f''(\xi)/2$ for some ξ); it is the coefficient of x^2 , that is, $a_2 = c_2$.
 $a_2 > 0 \iff q$ has a minimum.

From $q'(\bar{x}) = a_1 + a_2(2\bar{x} - x_1 - x_2) = 0$:

$$\bar{x} = \frac{x_1 + x_2}{2} - \frac{a_1}{2a_2}$$

In the example: $a_1 = -24$, $a_2 = 10$,
 $\bar{x} = 1.5 + 1.2 = 2.7$ — the same.

Algorithm: successive quadratic interpolation

Powell: x_1 , step Δ , ε_x , ε_f

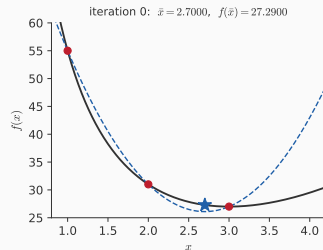
```
1: set  $x_2 = x_1 + \Delta$ ; evaluate  $f_1, f_2$ 
2: if  $f_1 > f_2$ : set  $x_3 = x_1 + 2\Delta$ ; else set  $x_3 = x_1 - \Delta$ ; evaluate  $f_3$ 
3: loop
4:   sort the three points:  $x_1 < x_2 < x_3$ 
5:   set  $F_{\min} = \min(f_1, f_2, f_3)$  and  $X_{\min}$  to the corresponding  $x_i$ 
6:   compute  $a_1, a_2$ ; if  $a_2 \leq 0$ : reflect the worst through the best;
   repeat
7:   set  $\bar{x} = \frac{x_1 + x_2}{2} - \frac{a_1}{2a_2}$ ; evaluate  $f(\bar{x})$ 
8:   if  $|F_{\min} - f(\bar{x})| \leq \varepsilon_f$  and  $|X_{\min} - \bar{x}| \leq \varepsilon_x$  then
9:     return the best of the four points
10:  end if
11:  keep the best point and the two that bracket it; if they do
    not exist, the three best
12: end loop
```

- Steps 1–2: the third point is placed on the side where f decreases, to try to *bracket* the minimum from the start.
- If x_2 is the best and is *bracketed* by two worse points, the parabola is convex and its minimum lies inside (x_1, x_3) .
- Cost: 3 initial evaluations + **1 per iteration** (+1 for each reflection).
- **Local superlinear** convergence (order ≈ 1.3 , under regularity conditions), without derivatives.

Example: $f(x) = x^2 + 54/x$, $x_1 = 1$, $\Delta = 1$

k	x_1	x_2	x_3	f_1	f_2	f_3	a_1	a_2	$\bar{x}$	$f(\bar{x})$
0	1.0000	2.0000	3.0000	55.0000	31.0000	27.0000	-24.000	10.000	2.7000	27.2900

Start: $x_2 = 2$; $f(1) = 55 > f(2) = 31 \Rightarrow x_3 = 1 + 2\Delta = 3$, $f(3) = 27$.

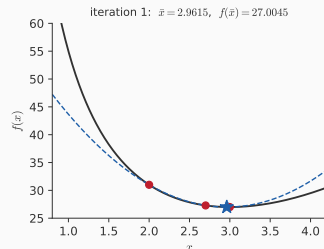


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1	2.0000	2.7000	3.0000	31.0000	27.2900	27.0000	-5.300	4.333	2.9615	27.0045

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After it. 0: best point $x = 3$; with no point on its right to bracket it, the three best are kept: 2, 2.7, 3.

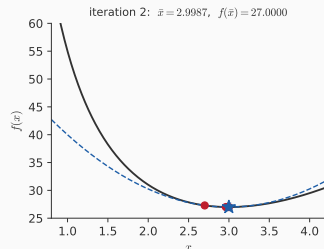


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1	2.0000	2.7000	3.0000	31.0000	27.2900	27.0000	-5.300	4.333	2.9615	27.0045
2	2.7000	2.9615	3.0000	27.2900	27.0045	27.0000	-1.092	3.251	2.9987	27.0000

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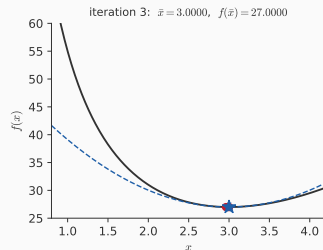
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2	2.7000	2.9615	3.0000	27.2900	27.0045	27.0000	-1.092	3.251	2.9987	27.0000
3	2.9615	2.9987	3.0000	27.0045	27.0000	27.0000	-0.120	3.027	3.0000	27.0000

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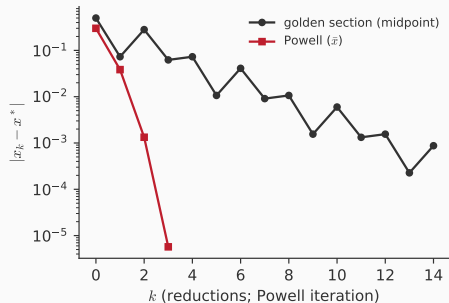
After it. 0: best point $x = 3$; with no point on its right to bracket it, the three best are kept: 2, 2.7, 3.

End (it. 3, the 4th): $|F_{\min} - f(\bar{x})| < 10^{-3}$ and $|X_{\min} - \bar{x}| < 10^{-3}$.

$x^* = 3$, $f(x^*) = 27$; 7 evaluations of f .



Golden section vs. Powell on the same problem



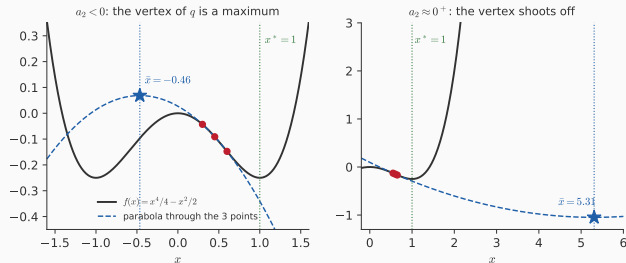
- Golden section on $[1, 4]$: interval $\times 0.618$ per reduction; the error of the midpoint oscillates and falls below 10^{-3} after 13 reductions (guarantee $L_k/2: 16$).
- Powell: error $0.3 \rightarrow 0.04 \rightarrow 0.0013 \rightarrow 6 \cdot 10^{-6}$ — 4 iterations, 7 evaluations.

But

With f unimodal, golden section always converges; quadratic interpolation may give $a_2 \leq 0$ or jump far away (next page). In practice they are combined (fminbnd).

Note — finite precision (bridge to 2.6): no method that only compares values of f can distinguish differences below $\varepsilon_m |f^*|$; here this limits $|x - x^*|$ to about $\sqrt{2\varepsilon_m |f^*| / f''(x^*)} \approx 4 \cdot 10^{-8}$.

When the parabola does not work



- $a_2 < 0$ (left, points in a concave region): q has a *maximum*; $\bar{x} = -0.46$ goes the wrong way.
- $a_2 \approx 0^+$ (right, points almost aligned near an inflection): the vertex *shoots off* to $\bar{x} = 5.31$, $f(\bar{x}) = 184$.
- Remedy: if $a_2 \leq 0$, do not compute $\bar{x}$ — **reflect the worst point through the best** (moving in the direction in which f decreases); and, as soon as possible, keep the best point **bracketed** between two worse ones: then $a_2 > 0$ and $\bar{x}$ falls inside the interval. The case of small $a_2 > 0$ is solved only by bracketing; `fminbnd` (Brent) also limits the step.

On the computer

The core of one iteration, in MATLAB:

```
[X, i] = sort(X);  F = F(i);           %  $x_1 < x_2 < x_3$ 
a1 = (F(2)-F(1)) / (X(2)-X(1));
a2 = ((F(3)-F(1))/(X(3)-X(1)) - a1) / (X(3)-X(2));
if a2 <= 0          % parabola with no minimum:
    % reflect the worst point through the best and repeat
else
    xb = (X(1)+X(2))/2 - a1/(2*a2);
    fb = f(xb);           % 1 evaluation
    % stop if xb and fb change little; otherwise keep
    % the best point and the two that bracket it
end
```

The full function (PowellQuadratic, with the safeguards and info.nfev) is in the course companion code.

In the example ($x_1 = 1$, $\Delta = 1$, tolerances 10^{-3}): 4 iterations, no reflections:
 $n_f = 3 + 4 = 7$.

Safeguard: with $a_2 \leq 0$ the best point is at an end. Reflecting the worst through the best, $2X_{\min} - x_{\text{worst}}$, generates a point on the other side of the best, outside the trio, in the direction in which the observed values indicate descent. In class: the three best and, with $a_2 \leq 0$, a random point; here, as in Deb (Step 6), bracket and reflect (deterministic). Same result in the class exercises.

MATLAB's `fminbnd` (Brent) alternates between parabolic steps and golden section steps: the speed of one, the guarantee of the other.

KEY TAKEAWAYS — 2.3

1. Golden section uses only the *order* of the values of f ; successive quadratic interpolation also uses their **magnitude**: it fits a parabola to three points and evaluates f at its minimum.
2. $\bar{x} = \frac{x_1 + x_2}{2} - \frac{a_1}{2a_2}$, with a_2 (curvature) > 0 . On a strictly convex quadratic, a single iteration.
3. Model $\rightarrow$ prediction $\rightarrow$ new evaluation: **one** evaluation per iteration and, near the minimum, locally and under suitable assumptions, superlinear convergence (order ≈ 1.3).
4. On its own it has no guarantee: $a_2 \leq 0$ or almost aligned points require safeguards (reflection, bracketing); in practice it is combined with golden section (`fminbnd`).

Next (2.4): what if we have f' ? At an interior minimum $f'(x^*) = 0$. As in order 0, we start by **comparing**: the *sign* of f' tells on which side the zero is — bisection.

References Full list: *Course bibliography*, at the end of the course PDF.

- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 2.
- Fletcher, R., *Practical Methods of Optimization*, 2nd ed., Wiley, 1987 — line search by interpolation.

Resources

- Companion code: `PowellQuadratic` (safeguards, `info.nfev`). MATLAB: `fminbnd` (Brent).